FlowSight

NETWORK ANOMALY DETECTION

Isolation Forest vs Autoencoder

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AI in Cybersecurity based on NVIDIA Morpheus

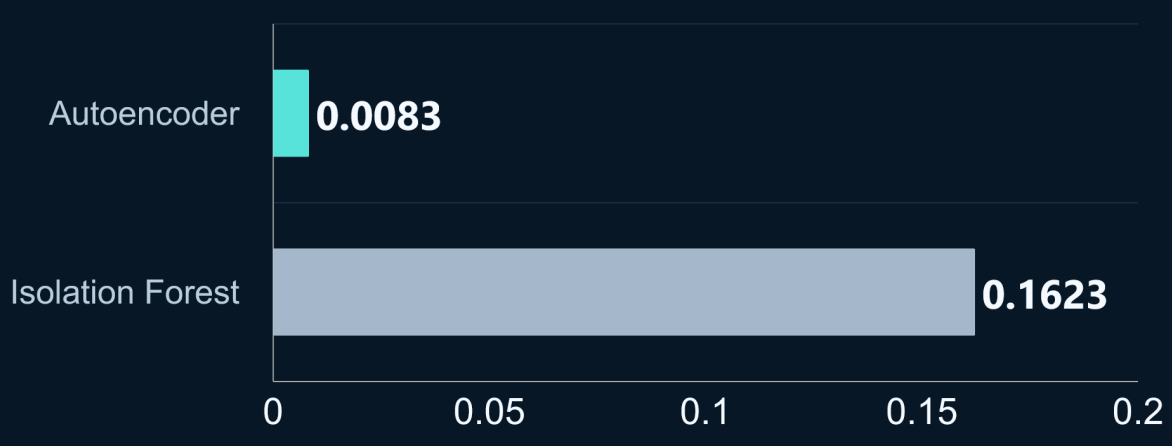


19.6×
Lower Autoencoder
P95 inference latency

400 / 400
Anomalies detected
by each model

104 vs 108
False alerts
Isolation Forest vs Autoencoder

Speed and detection quality



P95 latency in milliseconds. Lower is faster.
Inference only: preprocessing and file I/O excluded.

Metric	Isolation Forest	Autoencoder
Throughput (flows/s)	8,060.4	113,241.3
F1 score	88.50%	88.11%
False-positive rate	2.26%	2.35%

A controlled experiment

3,000 normal training flows
5,000 labeled test flows
400 anomalies / 4,600 normal

18 features from EVE JSON flows.
Shared scaling and event order.
Seed 42. Batch size 1.
Calibration contamination: 0.02.

Two NumPy models

Isolation Forest
100 trees, 256 samples per tree.

Autoencoder
18–6–18 network, tanh encoder,
linear decoder, Adam, 60 epochs.

What the experiment tells us

Autoencoder was faster in this synthetic CPU run. Both models detected every anomaly, but about one in five alerts was false. Real traffic and repeated runs are needed before deployment.

SOC and email scenarios demonstrate manual analyst review. ATT&CK candidates include T1046 and T1498.001, subject to contextual evidence. Morpheus, RAPIDS, Triton and GPU results remain future work.

SCAN FOR LIVE DEMO